

ER Diagram to Relational Schema

Database Management Systems

Learning Objectives

- Understand mapping rules
- Convert ER diagrams into relations
- Handle entities, relationships and attributes

Step 1: Strong Entities

Rule:

- Create one table for each strong entity.
- Primary key becomes PK.

Example:

STUDENT(StudentID, Name, Dept)

Step 2: Weak Entities

Rule:

- Create separate table.
- Include owner's PK as FK.

Example:

DEPENDENT(EmpID, DependentName, Age)

PK=(EmpID, DependentName)

Step 3: 1:1 Relationship

Example:

PERSON — Has — PASSPORT

Schema:

PERSON(PersonID, Name, PassportNo FK)

or

PASSPORT(PassportNo, PersonID FK)

Step 4: 1:N Relationship

Example:

DEPARTMENT(DeptID, DeptName)

EMPLOYEE(EmpID, Name, DeptID FK)

FK is stored on the 'many' side.

Step 5: M:N Relationship

Example:

STUDENT $\leftrightarrow$ COURSE

Create:

ENROLLMENT(StudentID FK, CourseID FK, Semester, Grade)

Composite PK=(StudentID, CourseID)

Step 6: Attributes

- Composite → split into simple attributes.

Address → Street, City, State

- Multivalued → new table

PHONE(StudentID, PhoneNo)

- Derived → usually not stored

Age from DOB